

AI61201: Visual Computing with AI/ML (Autumn 2025)

Programming Assignment 2: Image Transformations in Spatial Domain (20 Marks)

Due Date: September 03 (by 9 PM IST)

Instructions: Complete the four tasks given taking the following points into consideration:

- Vectorized implementations are possible for most operations, and using for or while loops to access individual pixels is strongly discouraged due to its inefficiency. You can use NumPy to perform the mathematical operations.
- You do not need to implement the convolution operations from scratch. However, you must design the filters from scratch wherever applicable.

Task 1: Image Exposure Check (3 Marks)

Write a function to check whether an image is overexposed, underexposed or correctly exposed. Classify the given set of images into one of these exposure types. Use `VC_a.jpg`, `VC_b.jpg`, `VC_c.jpg`, `VC_d.jpg`, `VC_e.jpg`, `VC_f.jpg` as inputs.

Task 2: Separable Gaussian Filtering (4 Marks)

- Implement an isotropic Gaussian filter with $\sigma = 11$ in a separable manner.
- Choose an appropriate kernel size for the given value of σ .
- Perform separable filtering with this filter kernel to denoise the given image (`noisy_image.jpg`)
- Compare the results of denoising with the 2D Gaussian filtering demonstrated in the tutorial in terms of visual results and computation time.

Task 3: Texture Classification Using Gabor Filtering (10 Marks)

- 3.a (3 marks)
 - Design a Gabor filter bank from scratch at different orientations (0, 45, 90, 135 (in degrees)), frequencies (0.25, 0.125, 0.0625), scales (using sigma values of 2, 4, 8) and aspect ratios (using gamma values of 0.5 and 1.0).
- 3.b (6 marks)
 - Load texture images for 3 textures (brick, gravel and grass) from `skimage.data` and display them.
 - Compute the response of the Gabor filter bank on the three images

- Show the level sets of a few of the filters from the Gabor filter bank and their responses to the three images.
- Compute the features of each of the three images as the mean and variance of each filter's response, i.e. if your filter bank has N filters, each image will have $2N$ corresponding features.
- Using the set of Gabor features computed on the three sample images and the corresponding Gabor features computed on the test images, classify each test image as brick, gravel or grass.
Use *img1.jpg*, *img2.jpg*, *img3.jpg*, *img4.jpg*, *img5.jpg*, *img6.jpg* as test images inputs
- 3.c (2 marks)
 - Perform the same experiment using OpenCV's implementation of Gabor filters with the same settings for the parameters.
 - Do the assigned categories for the test images match between your custom implementation of Gabor filters and OpenCV's implementation?

Task 4: Design of Sobel Operator (3 Marks)

Design a Sobel operator to detect 45 degree edges. Show the response of this filter to detect the 45 degree edges of the given *tower.jpg* image.

The inputs for all tasks have been included in the VisualComputing_Assignment2.zip uploaded with the questions

Submission Guidelines

1. *The content that you submit must be your individual work.*
2. *Submit your code in .py as well as in .ipynb file format. Both these file submissions are required to receive credit for this assignment*
3. *Ensure your code is well-commented and easy to follow. You can write your answers and explanations using text cells in the jupyter notebook files wherever required. For example, in Task 4, you need to state which bit plane you have chosen to replace, along with proper justification for your choice.*
4. *The files should be named as "<roll_number>_assignment_2". For example, if your roll number is 23AI91R01, the code the required file names will be 23AI91R01_assignment_2.py, 23AI91R01_assignment_2.ipynb. You should place all these files within a single zip file and upload it to Moodle as 23AI91R01_assignment_1.zip.*
5. *All submissions must be made through Moodle before the deadline. The submission portal will close at the specified time, and submissions via email would not be accepted.*

TA for this assignment: Shubhi Yadav

*If you have any queries regarding Assignment 2, please email at
shubhiyadav.24@kgpian.iitkgp.ac.in*